



UNIVERSITY OF MORATUWA
Faculty of Information Technology

B.Sc. (Hons) in Information Technology
Level 1 – Semester 2 Examination

IN1501 - Data Communication

Time Allowed: 3 Hours

May 2023

ADDITIONAL MATERIAL

None

INSTRUCTIONS TO CANDIDATES

1. This paper contains four (4) pages (including this page).
2. This paper comprises two (2) sections, namely Section A and Section B.
3. Section A contains five (5) short answer questions and Section B contains three (3) essay-type questions.
4. The total marks obtainable for this examination is 100. The marks assigned for each Question & Sections thereof are included in brackets.
5. This examination accounts for 60% of the module assessment.
6. This is a closed-book examination.
7. All the workings need to be shown very clearly.
8. Answer **ALL** Questions in both Sections A & B.
9. Answer Sections A and B in **separate** booklets.
10. **"Non-programmable"** scientific calculators are allowed.

Continued...

Section A

Answer all questions, clearly showing the workings.

(All complete and correct answers will carry 5 Marks each)

Q1 : A broadcast-quality video signal made up of 640x360 pixels per frame has been transmitted over a channel at the rate of 25 frames per second. Each pixel of the signal has been coded using 12 bits per primary colour of red, blue and green for the purpose of transmission. It was observed that when the channel was in operation for a continuous period of six hours from 0900 to 1500 hours, 5023 bits have been received in error.

- (i) Calculate the total number of bits required to create a complete frame of picture.
- (ii) Calculate the total number of bits transmitted during the above said period.
- (iii) Calculate the bit error rate of the above channel. (*Note: State the answer in standard form.*)

Q2 : A line-of-sight direct wireless communication channel has been established between two cities separated by 15 km. In order to receive signals with sufficient power, it has been identified that both ends of the system need to be fitted with antennas of gain 40 dB each and the frequency of the signal used is 5GHz. What are the cross-sectional areas (aperture) of the antennas used?

(Assume the speed of light in free space to be 3×10^8 m/s)

Q3 : Draw both the amplitude and phase spectra of the resulting signal $S_1 S_2$, when the signals $S_1 = -10 + 3\sin(75\pi t) + 7\sin(225\pi t - \pi/2)$ and $S_2 = 3\sin(5\pi t) + 6\sin(8\pi t + \pi/4)$ are multiplied with each other.

Q4 : A communication satellite is suspended in a geostationary orbit in the sky. The following additional information has been given to you.

Speed of the satellite $v = \sqrt{\frac{GM_E}{h_o}}$ where,

h_o is the altitude of the satellite from the earth's centre,

$G = 6.673 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ is the gravitational constant

$M_E = 5.9 \times 10^{24} \text{ kg}$ is the earth's mass and

$r = 6375 \text{ km}$ is the earth's radius at the equator

- (i) Calculate the altitude (height) of the satellite from the earth's surface.
- (ii) Calculate the average delay of a signal transmitted/received between two earth stations on earth neglecting processing delay by the satellite.

Q5 : A large corporation consisting of 12 offices in different parts of the country wants to implement a fully connected mesh network to connect all its offices together for seamless communication. Showing all your workings, calculate the total number of network links needed to implement this network.

End of Section A

Section B

Answer all questions.

Question One

- (a) Name and briefly describe the five generations of mobile communication. (5 Marks)
- (b) With the aid of a suitable example, describe the signal interference problem in mobile communication and possible solutions for overcoming such interference problem. (5 Marks)
- (c) With reference to cellular mobile communication, explain the terms "frequency reuse" and "frequency reuse pattern". (5 Marks)
- (d) For a cellular network using hexagonal cells,
- Prove that the co-channel reuse ratio is $(3N)^{1/2}$ where $N = i^2 + ij + j^2$.
 - Determine the number of cells in a cluster for the following values of the shift parameters i and j in a regular hexagon geometry pattern:
 - $i = 2$ and $j = 4$
 - $i = 3$ and $j = 3$
- (10 Marks)

Question Two

- (a) With the aid of suitable examples, briefly define the terms; odd symmetry, even symmetry and half-wave symmetry with respect to the shape of signals. (6 Marks)
- (b) Compute the Fourier transform of the following signal f shown in Figure Q2.

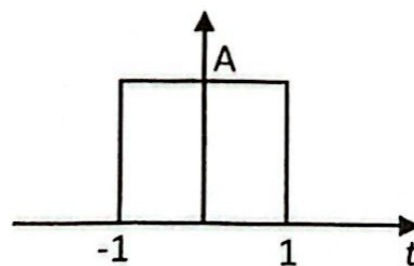


Figure Q2

- (15 Marks)
- (c) Plot the frequency and phase spectra of the above signal. (4 Marks)

Question Three

- (a) Name and briefly describe the three types of dispersion that may occur in an optical fiber communication system.

(6 Marks)

- (b) Briefly describe the terms; "*critical angle*" and "*total internal reflection*" with reference to the transmission of signals in an optical fiber.

(4 Marks)

- (c) A step-index optical fiber with the following parameters has been used to connect two locations within a city.

Refractive index of the core = 1.5

Refractive index of the cladding = 1.45

Diameter of the core = 50nm

Distance between the two locations = 5km

Speed of light in free space = 3.0×10^8 m/s

Compute the following:

- (i) Critical angle of the signal that reflects at the edge between the core and cladding.
- (ii) If the *Numerical Aperture (NA)* of a fiber is defined as the sine of the largest angle (outside the fiber) an incident ray can have for total internal reflectance in the core, Calculate the Numerical Aperture for this step-index fiber.
- (iii) Calculate the delay between the fastest and slowest optical signals travelling in this fiber between the above-said locations.

(15 Marks)

End of Section B

End of Paper